Complete Weierstrass elliptic function solutions for coherent couplers and their relation to degenerate four-wave mixing

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Abstract

Complete analytic solutions for the coherent coupler with arbitrary propagation constants and self- and cross-phase modulation coefficients are presented in terms of Weierstrass elliptic $\wp$, ζ, and σ functions, providing the full complex envelopes for both modes under arbitrary initial conditions. Jensen's coupler emerges as a special case of the general system. The solutions take the form of ratios of Weierstrass σ functions multiplied by a phase factor whose argument is a logarithm of σ ratios, introducing an essential singularity that is shown to be removable by a gauge transformation. A projection from the three-mode degenerate four-wave mixing system onto the two-mode coupler is identified, under which the solutions in the degenerate system become single-valued meromorphic Kronecker theta functions. This connection establishes the coherent coupler as a reduction of a broader class of integrable parametric processes and opens a pathway to leveraging known expansions of Kronecker theta functions for further analysis of nonlinear coupler dynamics.

1 Introduction

The coherent coupler is one of the canonical nonlinear optical systems, describing the exchange of optical power between two evanescently coupled waveguide modes under the Kerr nonlinearity. In the seminal work of Jensen [1], the coupled-mode equations were analysed to reveal power-dependent switching behaviour, establishing the nonlinear coupler as a prototype device for all-optical signal processing. Despite this long history, closed-form analytic solutions for the full complex envelopes of both modes have not previously been reported for the general system with arbitrary propagation constants and distinct self- and cross-phase modulation coefficients.

Most existing analytic treatments of the coupler follow the approach of Jensen and decompose the dynamics into modal powers and relative phase, leading to a single first-order equation for the fractional power imbalance that can be inverted in terms of Jacobi elliptic functions [1]. While this yields the power evolution exactly, it does not directly provide the complex envelopes, and the analysis is typically restricted to the symmetric case in which the two modes share identical propagation constants and nonlinear coefficients. The generalisation to asymmetric couplers introduces additional parameters that complicate the Jacobi elliptic function approach, and solutions for the complex envelopes in this general setting have remained elusive.

The present work derives complete solutions for the complex envelopes of the general asymmetric coupler using Weierstrass elliptic functions. The Weierstrass formulation provides a natural solution pathway: the coupled equations are recast as logarithmic derivatives, which directly yield Weierstrass ζ functions, and integration produces solutions expressed as ratios of Weierstrass σ functions. This progression through the Weierstrass hierarchy — from $\wp$ to ζ to σ — emerges organically from the structure of the complex envelope equations. The resulting solutions contain a phase factor whose argument is a logarithm of σ ratios, which introduces an essential singularity; we show that a gauge transformation removes this singularity, yielding a cleaner canonical form. The exact Weierstrass elliptic function solutions for the general asymmetric coupler constitute the first main result of this paper.

The second main result concerns the relationship between the coherent coupler and degenerate four-wave mixing (FWM). Degenerate FWM is a three-mode parametric process in which two pump photons are converted to a signal-idler pair sharing a common frequency, governed by a system of coupled-mode equations that is known to be integrable [2, 3]. We identify a projection from this three-mode system onto the two-mode coupler equations, establishing that the coupler is a reduction of the degenerate FWM system. Under this projection, the solutions in the degenerate system take the form of single-valued meromorphic Kronecker theta functions [4], a class of functions that appears across several integrable nonlinear optical systems including wave mixing in quadratic media and polarisation dynamics. This connection situates the coherent coupler within a broader integrable structure and suggests that the mathematical framework developed here may extend to other parametric processes.

The paper is organised as follows. Section 2 introduces the general coupled-mode equations and the relationship to Jensen's coupler. Section 3 derives the Weierstrass elliptic function solutions and the gauge transformation. Section 4 establishes the projection from degenerate FWM and the Kronecker theta function form of the solutions. Section 5 presents numerical validation. Section 6 concludes with a discussion of implications and possible extensions.

2 Jensen's coherent coupler system

In this section we recall the coherent coupler system from Jensen's classic paper [1] that forms the starting point for our study. Jensesn's system of ordinary differential equations shown in (2.1) describes propagation in the quasi-continuous-wave limit, i.e., when temporal envelope variations are negligible compared to the spatial evolution along the propagation direction, allowing time derivatives to be neglected. Physically, the equations govern the evolution of optical waves travelling in two wave guides put in close proximity to each other that exchange energy through coherent coupling of their evanescent fields. This interaction is modelled by the linear coupling terms parameterised by Ω_2. Each wave also experiences self-phase modulation (SPM), where its own intensity induces a phase shift on itself, and cross-phase modulation (XPM), where the intensities of the other wave induce additional phase shifts:

$$\begin{aligned}\frac{dA_1}{dz} &= i\left(|A_1|^2\Omega_3 + 2|A_2|^2\Omega_4\right)A_1 + i\Omega_1 A_1 + i\Omega_2 A_2,\\ \frac{dA_2}{dz} &= i\left(2|A_1|^2\Omega_4 + |A_2|^2\Omega_3\right)A_2 + i\Omega_2 A_1 + i\Omega_1 A_2,\end{aligned} \tag{2.1}$$

where:

- A_j with $j = 1, 2$ are the slowly-varying complex field amplitudes
- z is the propagation distance
- Ω_1 are the propagation constants
- Ω_2 is the linear coupling coefficient
- Ω_3 is the self-phase modulation (SPM) coefficient
- Ω_4 is the cross-phase modulation (XPM) coefficient

We refer the reader to (2.1) for details on how to express the Ω_j coefficients in terms of material parameters and overlap integrals. Jensen states that the waveguides in (2.1) were taken to be identical in order to simplify analysis and obtain solutions in standard elliptic integrals, hence the symmetry in the parameters in the two expressions. In this work however we analytically solve in full a generalised system that does not enforce such symmetry in the Ω_j parameters. Jensen's system emerges as a special case. We present the generalised system in the next section.

3 The generalised coherent coupler

In this section we normalise and generalise Jensen's system given in (2.1) to arbitrary coefficients to broaden the scope of applicability. We will use the same notation that we used in [3] for a general FWM system, the only difference being here that there are two modes not four, so mode index j takes values $1, 2$, whereas in the FWM case $j = 1, 2, 3, 4$. The system that we will consider is thus:

$$\begin{aligned}\frac{d}{dz}u_j(z) &= -\left(a_j + \sum_{k=1}^{4} a_{j,k}\, u_k v_k\right)u_j + \prod_{k=1,k\neq j}^{2} v_k,\\ \frac{d}{dz}v_j(z) &= \left(a_j + \sum_{k=1}^{4} a_{j,k}\, u_k v_k\right)v_j - \prod_{k=1,k\neq j}^{2} u_k,\end{aligned} \tag{3.1}$$

where in we refer to parameters $a_j, a_{j,k} \in \mathbb{C}$ as propagation constants and phase modulation parameters, respectively, and without loss of generality we take $a_{2,1} = a_{1,2}$. As there are only two modes, the product of modes governing wave mixing in (3.1) only contains one term and that term provides the linear coupling. Jensen's system in (2.1) is recovered from (3.1) by the following substitutions:

$$\begin{aligned}
a_1 &= -a_2 = -\frac{i\Omega_1}{\Omega_2}, \\
a_{1,1} &= a_{2,2} = \frac{i\Omega_3}{\Omega_2}, \\
a_{1,2} &= a_{2,1} = -\frac{2i\Omega_4}{\Omega_2}, \\
u_1(z) &= A_1\left(\frac{z}{\Omega_2}\right)e^{-\frac{i\pi}{4}}, \\
u_2(z) &= A_2\left(\frac{z}{\Omega_2}\right)e^{\frac{i\pi}{4}}, \\
v_1(z) &= -A_1^*\left(\frac{z}{\Omega_2}\right)e^{\frac{i\pi}{4}}, \\
v_2(z) &= A_2^*\left(\frac{z}{\Omega_2}\right)e^{-\frac{i\pi}{4}}, \qquad (3.2)
\end{aligned}$$

where A^* denotes the complex conjugate, and where having made the subsitution (3.2) in (3.1) one rescales the length variable to a new variable z' such that $z = \Omega_2 z'$. Therefore, Jensen's system in (2.1) is a special case of the more general system in (3.1).

In the next section we provide the complete analytic solution for the generalised system in (3.1). Following our approach in [3], from this point on we do not assume that u_j and v_j are necessarily complex conjugates, but we will show in the following section that they are Hamiltonian conjugates with physical systems such as Jensen's being a particular realisation of a more general system. In all that follows we refer to u_j, v_j as modes, which we use as a general term for components of the coupled system. We refer to their product $u_j v_j$ as the modal power; while this quantity is not necessarily real-valued, the conceptual analogy is useful.

4 Conserved quantities

In this section we identify conserved quantities of the system. The approach and notation follows that of [3] adjusted for two modes instead of four. The system in (3.1) can be formulated as the following canonical Hamiltonian system:

$$H(u_1,u_2,v_1,v_2) = -\sum_{j=1}^{2} a_j\, u_j v_j - \frac{1}{2}\sum_{j,k=1}^{2} a_{j,k}\, u_j v_j u_k v_k + \prod_{j=1}^{2} u_j + \prod_{j=1}^{2} v_j, \qquad (4.1)$$

$$\frac{d}{dz}u_j(z) = \frac{\partial H}{\partial v_j}, \qquad (4.2)$$

$$\frac{d}{dz}v_j(z) = -\frac{\partial H}{\partial u_j}, \qquad (4.3)$$

$$\frac{d}{dz}H = 0 \qquad (4.4)$$

The conservation of H is to be expected as a consequence of (4.1) lacking an explicit z dependence. The pair u_j, v_j are Hamiltonian conjugates, and each pair represents one of two degrees of freedom in the system. The modal power $u_j v_j$ evolves according to:

$$\frac{d}{dz}u_j v_j = -\prod_{k=1}^{2} u_k + \prod_{k=1}^{2} v_k \qquad (4.5)$$

As the right hand side of (4.5) is the same for all j, we may define constants γ_j and function $\rho(z)$ such that:

$$u_j(z)v_j(z) = \gamma_j - \rho(z) \qquad (4.6)$$

from which it follows that there is an intermodal power conservation law of the form:

$$u_1(z)v_1(z) - u_2(z)v_2(z) = \gamma_1 - \gamma_2. \qquad (4.7)$$

The system has two degrees of freedom and two conserved quantities which is a requirement for integrability in the Liouville-Arnold sense [5]. The γ_j constants can be determined from (4.7) and initial conditions, however, this provides one equation for two unknowns and thus a choice is available for normalisation. We choose to impose the constraint that:

$$\sum_{j=1}^{2} \gamma_j = 0, \tag{4.8}$$

after which $\rho(z)$ is minus the mean modal power and γ_j is the constant difference between a modes power and the mean.

5 Solutions for modal powers in terms of Weierstrass $\wp$ elliptic functions

In this section we give analytic solutions for modal power. As in the previous section, the approach and notation closely follows that of [3] adjusted for two modes instead of four. The fundamental Weierstrass elliptic function theory used in this and subsequent sections can be found in [6]. In order to obtain elliptic function solutions in wave mixing dynamics, the key observation is that the derivative of the modal power in (4.5) is proportional to the difference of the two wave mixing product terms, and that the Hamiltonian in (4.1) contains their sum. We then note the simple but important identity:

$$\left(\prod_{k=1}^{2} u_k - \prod_{k=1}^{2} v_k\right)^2 - \left(\prod_{k=1}^{2} u_k + \prod_{k=1}^{2} v_k\right)^2 = -4\prod_{j=1}^{2} u_j v_j \tag{5.1}$$

which enables us to square (4.5) and replace wave mixing terms with phase modulation terms, i.e., monomials of $u_j v_j$ which can all be expressed in terms of $\rho(z)$ via (4.6). To proceed in this manner, let us introduce the function Q which represents the phase modulation part of the Hamiltonian and is thus itself expressible as a polynomial of $\rho(z)$:

$$\begin{aligned} Q(u_1(z)v_1(z), u_2(z)v_2(z)) &= a_0 + \sum_{j=1}^{2} a_j\, u_j v_j + \frac{1}{2}\sum_{j,k=1}^{2} a_{j,k}\, u_j v_j u_k v_k, \\ &= \sum_{l=0}^{2} b_l\, \rho\,(z)^l \end{aligned} \tag{5.2}$$

with $a_0 = H$ a renaming that indicates the role of H here as the constant term of the polynomial. Observe that squaring (4.5) and substituting (4.1), (4.6), (5.1), and (5.2) gives:

$$\left(\frac{d}{dz}\rho(z)\right)^2 = Q^2(\gamma_1 - \rho(z), \gamma_2 - \rho(z)) - 4\prod_{j=1}^{2}(\gamma_j - \rho(z)), \tag{5.3}$$

$$\left(\frac{d}{dz}\rho(z)\right)^2 = \left(\sum_{l=0}^{2} b_l\, \rho\,(z)^l\right)^2 - 4\prod_{j=1}^{2}(\gamma_j - \rho(z)), \tag{5.4}$$

$$\left(\frac{d}{dz}\rho(z)\right)^2 = \sum_{l=0}^{4} d_l\, \rho\,(z)^l, \tag{5.5}$$

$$\left(\frac{d}{dz}\rho(z)\right)^2 = d_4\prod_{l=1}^{4}(\rho\,(z) - \lambda_l), \tag{5.6}$$

where b_l, and d_l are given in terms of other parameters and initial conditions in Appendices A and B, and where λ_l are the roots of the quartic polynomial in $\rho(z)$, i.e.:

$$\sum_{k=0}^{4} d_k\, \lambda_l^k = 0 \tag{5.7}$$

Equation (5.6) is a differential equation for an elliptic function and by separating phase modulation and wave mixing terms in the Hamiltonian, and by exploiting intermodal power conservation, we were able to obtain it directly without inverting elliptic integrals first. To solve it, we now transform (5.6) from quartic to the standard cubic form of the Weierstrass $\wp$ function using the classical trick which can be conceptualised in three steps [6]:

$$\begin{array}{lrcl} \text{(1) shift by any root so RHS is 0 when } q(z)=0 & \rho(z) & = & q(z) + \lambda_1, \\ \text{(2) invert to reduce the quartic to a cubic} & s(z) & = & \dfrac{1}{q(z)}, \\ \text{(3) shift and scale to match Weierstrass coefficients} & w(z) & = & C_1 s(z) + C_0. \end{array} \tag{5.8}$$

Without loss of generality, we choose root λ_1. The procedure sketched in (5.8) is then implemented in the following single transformation:

$$\rho(z) = \lambda_1 + \frac{d_4}{-4w(z)\prod_{l=1}^{3}\Omega_l + \frac{d_4}{3}\sum_{l=1}^{3}\Omega_l}, \tag{5.9}$$

$$\Omega_l = \frac{1}{\lambda_{l+1}-\lambda_1},$$

$$\left(\frac{d}{dz}w(z)\right)^2 = 4\,w(z)^3 - g_2\,w(z) - g_3, \tag{5.10}$$

$$g_2 = d_0 d_4 - \frac{d_1 d_3}{4} + \frac{d_2^2}{12}, \tag{5.11}$$

$$g_3 = \frac{d_0 d_2 d_4}{6} - \frac{d_0 d_3^2}{16} - \frac{d_1^2 d_4}{16} + \frac{d_1 d_2 d_3}{48} - \frac{d_2^3}{216} \tag{5.12}$$

where the constants g_2 and g_3 are known as Weierstrass elliptic invariants. Equation (5.10) defines the Weierstrass elliptic $\wp$ function and the solution is:

$$w(z) = \wp(z - z_0, g_2, g_3) \tag{5.13}$$

where z_0 is a constant chosen to match initial conditions. This constant z_0, and others that we will now introduce, can be obtained by inverting $\wp$ using an elliptic integral [6]. As $\wp$ is an even function, it is necessary to also specify a corresponding condition for the derivative when inverting to fix the sign, i.e., to find z from known x, y we give conditions such as $\wp(z) = x, \wp'(z) = y$, where $\wp'$ is the derivative of $\wp$ known as Weierstrass p-prime. From elliptic function theory, the points obtained during such an inversion are determined modulo the period lattice of the doubly periodic $\wp$, which is a two-dimensional generalisation of saying they are only uniquely determined within a period of oscillation. In addition to z_0, we introduce two other types of special points that characterize the solution structure and enable the use of elliptic function identities to solve for the modes in Section 6. The point $z_0 + z_1$ corresponds to the pole of $\rho(z)$ where the denominator in (5.9) vanishes, and represents a singularity in the mean modal power. The points μ_j are the locations where individual modal powers vanish, $u_j(\mu_j)v_j(\mu_j) = 0$, and thus where $\rho(\mu_j) = \gamma_j$ from (4.6). These special points are defined implicitly through their Weierstrass function values:

$$\begin{aligned}
\wp(z_0) &= \frac{d_2}{12} + \frac{d_3\lambda_1}{4} + \frac{d_4\lambda_1^2}{2} + \frac{-d_1 - 2d_2\lambda_1 - 3d_3\lambda_1^2 - 4d_4\lambda_1^3}{4\left(-\rho(0)+\lambda_1\right)},\\
\wp'(z_0) &= \frac{\left(d_1 + 2d_2\lambda_1 + 3d_3\lambda_1^2 + 4d_4\lambda_1^3\right)}{4\left(\rho(0)-\lambda_1\right)^2}\left.\frac{d}{dz}\rho(z)\right|_{z=0},\\
\wp(z_1) &= \frac{d_2}{12} + \frac{d_3\lambda_1}{4} + \frac{d_4\lambda_1^2}{2},\\
\wp'(z_1) &= \frac{\left(-d_1 - 2d_2\lambda_1 - 3d_3\lambda_1^2 - 4d_4\lambda_1^3\right)b_2}{4},\\
\wp(\mu_j - z_0) &= \frac{d_2}{12} + \frac{d_3\lambda_1}{4} + \frac{d_4\lambda_1^2}{2} - \frac{-d_1 - 2d_2\lambda_1 - 3d_3\lambda_1^2 - 4d_4\lambda_1^3}{4\left(\gamma_j - \lambda_1\right)},\\
\wp'(\mu_j - z_0) &= -\frac{\left(b_0 + b_1\gamma_j + b_2\gamma_j^2\right)\left(d_1 + 2d_2\lambda_1 + 3d_3\lambda_1^2 + 4d_4\lambda_1^3\right)}{4\left(\gamma_j - \lambda_1\right)^2}.
\end{aligned} \tag{5.14}$$

The solutions for modal powers are then found from (4.6) and (5.9) to be:

$$\begin{aligned}
u_j(z)v_j(z) &= \rho(\mu_j) - \rho(z),\\
&= \frac{\wp'(z_1)}{b_2\left(\wp(z_1) - \wp(\mu_j - z_0)\right)} - \frac{\wp'(z_1)}{b_2\left(\wp(z_1) - \wp(z - z_0)\right)},\\
&= \frac{\wp'(z_1)}{b_2\left(\wp(z_1) - \wp(\mu_j - z_0)\right)}\frac{\left(\wp(z - z_0) - \wp(\mu_j - z_0)\right)}{\left(\wp(z - z_0) - \wp(z_1)\right)}.
\end{aligned} \tag{5.15}$$

6 Solutions for modes in terms of Weierstrass σ, ζ functions

Having obtained the modal power solutions in the previous section, we now proceed to find solutions for the individual modes $u_j(z)$ and $v_j(z)$ themselves. Direct substitution of (4.1), (4.5), (4.6), and (4.7) into (3.1) shows that the equations can be recast in terms of logarithmic derivatives:

$$\frac{\frac{\partial}{\partial z}u_j(z)}{u_j(z)} = \frac{1}{2}\frac{\rho'(z)-\rho'(\mu_j)}{\rho(z)-\rho(\mu_j)} + \rho(z)\Lambda_{1,j} + \Lambda_{0,j},$$

$$\frac{\frac{\partial}{\partial z}v_j(z)}{v_j(z)} = \frac{1}{2}\frac{\rho'(z)+\rho'(\mu_j)}{\rho(z)-\rho(\mu_j)} - \rho(z)\Lambda_{1,j} - \Lambda_{0,j}, \tag{6.1}$$

$$\Lambda_{0,j} = -a_j - \frac{\gamma_j}{4}\sum_{\substack{1\le k\le 2\\ 1\le l\le 2}} a_{k,l} - \sum_{l=1}^{2} a_{j,l}\gamma_l + \frac{1}{2}\sum_{k=1}^{2} a_{k,k}\gamma_k + \sum_{k=1}^{2}\frac{a_k}{2}, \tag{6.2}$$

$$\Lambda_{1,j} = \sum_{k=1}^{2} a_{j,k} - \frac{1}{4}\sum_{\substack{1\le k\le 2\\ 1\le l\le 2}} a_{k,l}, \tag{6.3}$$

$$\Lambda_{0,j} = -a_j - \frac{\gamma_j}{4}\sum_{k,l=1}^{4} a_{k,l} - \sum_{k=1}^{4} a_{j,k}\gamma_k + \frac{1}{2}\sum_{k=1}^{4}\gamma_k\sum_{l=1}^{4} a_{k,l} + \frac{1}{2}\sum_{k=1}^{4} a_k, \tag{6.4}$$

$$\Lambda_{1,j} = \sum_{k=1}^{4} a_{j,k} - \frac{1}{4}\sum_{k,l=1}^{4} a_{k,l}, \tag{6.5}$$

where the right-hand side is expressed purely in terms of ρ and its derivative ρ'. This logarithmic derivative form is advantageous because it can be integrated directly to yield solutions in terms of Weierstrass σ functions, as we now show.

We substitute the modal power solution (5.15) into (6.1) and apply the elliptic function identity:

$$\frac{\wp'(x,g_2,g_3)}{\wp(x,g_2,g_3)-\wp(y,g_2,g_3)} = \zeta(x+y,g_2,g_3) + \zeta(x-y,g_2,g_3) - 2\zeta(x,g_2,g_3) \tag{6.6}$$

which allows us to express the right-hand side in terms of Weierstrass ζ functions and constants:

$$\begin{aligned}\frac{\frac{\partial}{\partial z}u(z,\mu_j)}{u(z,\mu_j)} =& \frac{\left(\zeta(z-z_0+z_1)-2\zeta(z_1)-\zeta(z-z_0-z_1)\right)\Lambda_{1,j}}{b_2}\\ &+\zeta(z-2z_0+\mu_j) - \frac{\zeta(z-z_0-z_1)}{2} - \frac{\zeta(z-z_0+z_1)}{2}\\ &-\frac{\zeta(\mu_j-z_0-z_1)}{2} - \frac{\zeta(\mu_j-z_0+z_1)}{2} + \Lambda_{0,j} + \Lambda_{1,j}\lambda_1\\ \frac{\frac{\partial}{\partial z}v(z,\mu_j)}{v(z,\mu_j)} =& -\frac{\left(\zeta(z-z_0+z_1)-2\zeta(z_1)-\zeta(z-z_0-z_1)\right)\Lambda_{1,j}}{b_2}\\ &+\zeta(z-\mu_j) - \frac{\zeta(z-z_0-z_1)}{2} - \frac{\zeta(z-z_0+z_1)}{2}\\ &+\frac{\zeta(\mu_j-z_0-z_1)}{2} + \frac{\zeta(\mu_j-z_0+z_1)}{2} - \Lambda_{0,j} - \Lambda_{1,j}\lambda_1.\end{aligned} \tag{6.7}$$

Equations (6.7) can be integrated by noting that the Weierstrass ζ function is the logarithmic derivative of the Weierstrass σ function:

$$\zeta\left(z,g_2,g_3\right) = \frac{\partial}{\partial z}\log\left(\sigma\left(z,g_2,g_3\right)\right). \tag{6.8}$$

Performing the integration and taking exponentials gives the first main result of this paper, the complete analytic solutions for the complex mode envelopes u_j, v_j without decomposition into amplitude and phase:

$$u_j(z) = \frac{\alpha_j\sqrt{W_j}\,\sigma(z-2z_0+\mu_j)\exp\left(zr_{0,j} + \log\left(\frac{\sigma(z-z_0+z_1)}{\sigma(z-z_0-z_1)}\right)r_{1,j}\right)}{\sqrt{\wp(z_1)-\wp(z-z_0)}\sigma(\mu_j-z_0)\sigma(z-z_0)},$$

$$v_j(z) = \frac{\sqrt{W_j}\,\sigma(z-\mu_j)\exp\left(-zr_{0,j} - \log\left(\frac{\sigma(z-z_0+z_1)}{\sigma(z-z_0-z_1)}\right)r_{1,j}\right)}{\alpha_j\sqrt{\wp(z_1)-\wp(z-z_0)}\sigma(\mu_j-z_0)\sigma(z-z_0)} \tag{6.9}$$

where the branch of the logarithm is chosen continuously along the integration path from $z = 0$, α_j is the integration constant that can be fixed by initial conditions to capture any phase offset between a mode and its conjugate, and the other constants are:

$$W_j = \frac{\wp'(z_1)}{b_2\,(\wp(z_1) - \wp(\mu_j - z_0))}, \tag{6.10}$$

$$r_{0,j} = \Lambda_{0,j} + \Lambda_{1,j}\lambda_1 - \frac{2\zeta(z_1)\Lambda_{1,j}}{b_2} - \frac{\zeta(\mu_j - z_0 - z_1)}{2} - \frac{\zeta(\mu_j - z_0 + z_1)}{2}, \tag{6.11}$$

$$r_{1,j} = \frac{\Lambda_{1,j}}{b_2}, \tag{6.12}$$

and it can be shown that:

$$\sum_{j=1}^{2} r_{1,j} = 1. \tag{6.13}$$

These solutions are valid for all initial conditions without approximation.

We show in Appendix C that the Hamiltonian (4.1) can be expressed as the Frobenius–Stickelberger determinant, a classical identity relating products of Weierstrass σ functions to determinants of derivatives of $\wp$. This provides a deeper structural understanding of how the Hamiltonian is conserved and connects the four-wave mixing system to the broader theory of elliptic functions.

7 Transforming to canonical coordinates

In this section we present the second main result of this paper: a canonical coordinate system for four-wave mixing that substantially simplifies both the governing equations and their solutions. The structure of the analytic solutions in (6.9) naturally suggests this transformation. In particular, the presence of the factor $\sqrt{\wp(z_1) - \wp(z - z_0)}$ in the denominator motivates a local normalisation, while the z-dependent logarithmic terms weighted by $r_{1,j}$ suggest a gauge transformation to remove cross-phase modulation.

Remarkably, although the transformation depends explicitly on the propagation variable z, it leaves the coupled differential equations structurally invariant, preserving the characteristic four-wave mixing interaction terms while achieving substantial simplifications. In the canonical coordinates, cross-phase modulation is eliminated, the differential equations for modal powers reduce from quartic to cubic polynomials, system parameters are greatly reduced (and can be completely removed through rescaling), and the solutions become single-valued meromorphic Kronecker theta functions. At the same time, the transformation preserves the Hamiltonian structure, all intermodal power conservation laws, and intermodal power ratios. We accomplish this through a sequence of three transformations, which we now present in turn.

7.1 Removing cross-phase modulation with a gauge transform

To begin, let us consider a transform of the system in (3.1) of the following form:

$$\begin{aligned} u_j(z) &= \hat{u}_j(z)e^{-\phi_j(z)}, \\ v_j(z) &= \hat{v}_j(z)e^{\phi_j(z)}, \end{aligned} \tag{7.1}$$

$$\sum_{j=1}^{2} \phi_j(z) = 0, \tag{7.2}$$

where transforming the conjugate mode with the opposing phase leaves modal powers unchanged, and where the sum over ϕ_j being zero ensures we do not encounter any new terms appearing in the exponents of wave mixing terms. We make the following choice for ϕ_j composed of a part linear in z (labelled L) and a part nonlinear in z (labelled NL) (where L and NL are function labels not exponents):

$$N = 2, \tag{7.3}$$
$$\phi_j(z) = \phi_j^L(z) + \phi_j^{NL}(z) \tag{7.4}$$
$$\phi_j^L(z) = z\left(a_j - \frac{1}{N}\sum_{k=1}^N a_k - \frac{\gamma_j}{N}\sum_{l,k=1}^N a_{l,k}\right), \tag{7.5}$$
$$\phi_j^{NL}(z) = \sum_{k=1}^N\left(a_{j,k} - \frac{1}{N}\sum_{l=1}^N a_{l,k}\right)\int \hat{u}_k(z)\hat{v}_k(z)\,dz \tag{7.6}$$
$$\sum_{j=1}^N \phi_j(z) = \sum_{j=1}^N \phi_j^L(z) = \sum_{j=1}^N \phi_j^{NL}(z) = 0. \tag{7.7}$$

This transforms (3.1) into the following system:

$$\begin{aligned}\frac{\partial}{\partial z}\hat{u}_j &= (b_1 - 2b_2\hat{u}_j\hat{v}_j)\frac{\hat{u}_j}{N} + \prod_{k=1,k\neq j}^N \hat{v}_k \\ \frac{\partial}{\partial z}\hat{v}_j &= (-b_1 + 2b_2\hat{u}_j\hat{v}_j)\frac{\hat{v}_j}{N} - \prod_{k=1,k\neq j}^N \hat{u}_k\end{aligned} \tag{7.8}$$

for which the conserved canonical Hamiltonian is:

$$\hat{H} = \prod_{l=1}^N \hat{u}_l + \prod_{l=1}^N \hat{v}_l - \frac{1}{N}\sum_{l=1}^N\left(b_2\hat{u}_l^2\hat{v}_l^2 - b_1\hat{u}_l\hat{v}_l\right) \tag{7.9}$$

and the intermodal power conservation laws are unchanged such that $\hat{u}_j\hat{v}_j - \hat{u}_k\hat{v}_k = \gamma_j - \gamma_k$. To maintain the connection with the original coordinate system the value of $\hat{H}$ would be fixed at:

$$\hat{H} = -\frac{(\gamma_1-\gamma_2)^2 b_2}{4} + b_0. \tag{7.10}$$

Comparing (7.8) with the original system (3.1), we observe significant simplification. All modes now share a single propagation constant b_1/N and a single self-phase modulation coefficient $2b_2/N$, whilst all cross-phase modulation terms $a_{j,k}$ with $j \neq k$ have been eliminated. It should be noted that parameters b_1, b_2 depend on γ_j and thus, in the new coordinates parameters depend on mode power in the original coordinates and are not exclusively fiber parameters. In terms of the solutions to the system, the effect of the ϕ_j^{NL} transform is to remove the z dependent log terms weighted by $r_{1,j}$ in (6.9). The solutions are:

$$\begin{aligned}\hat{u}_j(z) &= \frac{\hat{\alpha}_j\sqrt{W_j}\sigma(z_1)\sigma(z-2z_0+\mu_j)e^{z\left(r_{0,j}-(-1)^j\beta\right)}}{\sigma(\mu_j-z_0)\sigma(z-z_0-z_1)}, \\ \hat{v}_j(z) &= \frac{\sqrt{W_j}\sigma(z_1)\sigma(z-\mu_j)e^{-z\left(r_{0,j}-(-1)^j\beta\right)}}{\hat{\alpha}_j\sigma(\mu_j-z_0)\sigma(z-z_0+z_1)},\end{aligned} \tag{7.11}$$

where:

$$\beta = \frac{(-r_{1,1}+r_{1,2})(-2\zeta(z_1)+b_2\gamma_1+b_2\lambda_1)}{2} - 2a_{1,2}\gamma_1 - a_2 - \frac{b_1}{2} \tag{7.12}$$

LOOK AT SIMPLIFYING beta r0j using LAM0

7.2 Projecting from degenerate FWM to the coherent coupler

In this section we will show how the coherent coupler system in (3.1) is essentially equivalent to a degenerate four-wave mixing (FWM) system that one finds in nonlinear fiber optics. To be more precise, we will show that there is a projection from the three mode degenerate FWM system to the two mode coherent coupler system.

To start, let us recall that in [3] we showed how quasi-continuous-wave FWM systems can be transformed into the following parameterless non-degenerate canonical Hamiltonian system of four modes $\tilde{u}_j(z), j = 1,2,3,4$, and their Hamiltonian conjugates $\tilde{v}_j(z)$ (note the tilde notation):

$$\tilde{H} = \sum_{l=1}^4\left(\frac{1}{8}\tilde{u}_l^2\tilde{v}_l^2 - \tilde{u}_l\tilde{v}_l\right) - \frac{1}{4}\left(\prod_{l=1}^4\tilde{u}_l + \prod_{l=1}^4\tilde{v}_l\right), \tag{7.13}$$

$$
\begin{aligned}
\frac{\partial}{\partial z}\tilde{u}_j &= \frac{\partial}{\partial \tilde{v}_j}\tilde{H} = -\left(1-\frac{\tilde{u}_j\tilde{v}_j}{4}\right)\tilde{u}_j - \frac{1}{4}\prod_{k=1,k\neq j}^{4}\tilde{v}_k,\\
\frac{\partial}{\partial z}\tilde{v}_j &= -\frac{\partial}{\partial \tilde{u}_j}\tilde{H} = \left(1-\frac{\tilde{u}_j\tilde{v}_j}{4}\right)\tilde{v}_j + \frac{1}{4}\prod_{k=1,k\neq j}^{4}\tilde{u}_k,
\end{aligned} \tag{7.14}
$$

which conserves the Hamiltonian $\tilde{H}$ and, for constants $\tilde{\gamma}_j$, three intermodal power differences (read as $j > k$ to avoid over counting):

$$
\tilde{u}_j(z)\tilde{v}_j(z) - \tilde{u}_k(z)\tilde{u}_k(z) = \tilde{\gamma}_j - \tilde{\gamma}_k. \tag{7.15}
$$

We will now consider a degeneration and rescaling of this FWM system defined by the following substitutions in the Hamiltonian in (7.13) (note the bar notation on the right and the tilde notation on the left):

$$
\begin{aligned}
\tilde{u}_1(z) &= \varsigma\sqrt{2}\chi\,\bar{u}_1(z),\\
\tilde{u}_2(z) &= \sqrt{2}\chi\,\bar{u}_2(z),\\
\tilde{u}_3(z) &= \chi\,\bar{u}_3(z),\\
\tilde{u}_4(z) &= \chi\,\bar{u}_3(z),\\
\tilde{v}_1(z) &= \varsigma\sqrt{2}\chi\,\bar{v}_1(z),\\
\tilde{v}_2(z) &= \sqrt{2}\chi\,\bar{v}_2(z),\\
\tilde{v}_3(z) &= \chi\,\bar{v}_3(z),\\
\tilde{v}_4(z) &= \chi\,\bar{v}_3(z),
\end{aligned} \tag{7.16}
$$

where $\varsigma = \pm 1$, χ is a free parameter, and $\tilde{u}_3$ and $\tilde{u}_4$ coalesce to $\bar{u}_3$ (likewise for the corresponding conjugates) to create degeneracy and reduce four degrees of freedom (modes) to three. We then divide the Hamiltonian by $2\chi^3$, which is equivalent to a change of the length variable such that $z \to 2\chi^3 z$ (check this), to obtain the degenerate FWM system that we will start from:

$$
\bar{H} = \frac{\tilde{H}}{2\chi^3} = -\frac{\varsigma\chi}{4}\left(\bar{u}_1\bar{u}_2\bar{u}_3^2 - \bar{v}_1\bar{v}_2\bar{v}_3^2\right) + \frac{\chi\bar{u}_1^2\bar{v}_1^2}{4} + \frac{\chi\bar{u}_2^2\bar{v}_2^2}{4} + \frac{\chi\bar{u}_3^2\bar{v}_3^2}{8} - \frac{1}{\chi}\sum_{j=1}^{3}\bar{u}_j\bar{v}_j, \tag{7.17}
$$

$$
\begin{aligned}
\frac{\partial}{\partial z}\bar{u}_j(z) &= \frac{\partial}{\partial \bar{v}_j}\bar{H},\\
\frac{\partial}{\partial z}\bar{v}_j(z) &= -\frac{\partial}{\partial \bar{u}_1}\bar{H},
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial}{\partial z}\bar{u}_1(z) &= -\frac{\bar{u}_1}{\chi} + \frac{\chi\bar{u}_1^2\bar{v}_1}{2} - \frac{\varsigma\chi\bar{v}_2\bar{v}_3^2}{4},\\
\frac{\partial}{\partial z}\bar{u}_2(z) &= -\frac{\bar{u}_2}{\chi} + \frac{\chi\bar{u}_2^2\bar{v}_2}{2} - \frac{\varsigma\chi\bar{v}_1\bar{v}_3^2}{4},\\
\frac{\partial}{\partial z}\bar{u}_3(z) &= -\frac{\bar{u}_3}{\chi} + \frac{\chi\bar{u}_3^2\bar{v}_3}{4} - \frac{\varsigma\chi\bar{v}_1\bar{v}_2\bar{v}_3}{2},\\
\frac{\partial}{\partial z}\bar{v}_1(z) &= \frac{\bar{v}_1}{\chi} - \frac{\chi\bar{v}_1^2\bar{u}_1}{2} + \frac{\varsigma\chi\bar{u}_2\bar{u}_3^2}{4},\\
\frac{\partial}{\partial z}\bar{v}_2(z) &= \frac{\bar{v}_2}{\chi} - \frac{\chi\bar{v}_2^2\bar{u}_2}{2} + \frac{\varsigma\chi\bar{u}_1\bar{u}_3^2}{4},\\
\frac{\partial}{\partial z}\bar{v}_3(z) &= \frac{\bar{v}_3}{\chi} - \frac{\chi\bar{v}_3^2\bar{u}_3}{4} + \frac{\varsigma\chi\bar{u}_1\bar{u}_2\bar{v}_3}{2},
\end{aligned} \tag{7.18}
$$

We will be showing that there is a projection from (7.18) to (3.1).

Let us define modes $\hat{u}_j$, $\hat{v}_j$ with $j = 1, 2$ (note the hat notation), in terms of $\bar{u}_k$, $\bar{v}_k$ modes with $k = 1, 2, 3$, as follows:

$$\begin{aligned}\hat{u}_1(z) &= \frac{\sqrt{2}\sqrt{\gamma_1 - \lambda_1}\bar{u}_1(z)e^{z\theta}}{\bar{v}_3(z)},\\ \hat{u}_2(z) &= \frac{\sqrt{2}\sqrt{\gamma_2 - \lambda_1}\bar{u}_2(z)e^{-z\theta}}{\bar{v}_3(z)},\\ \hat{v}_1(z) &= \frac{\sqrt{2}\sqrt{\gamma_1 - \lambda_1}\bar{v}_1(z)e^{-z\theta}}{\bar{u}_3(z)},\\ \hat{v}_2(z) &= \frac{\sqrt{2}\sqrt{\gamma_2 - \lambda_1}\bar{v}_2(z)e^{z\theta}}{\bar{u}_3(z)},\end{aligned} \tag{7.19}$$

where:

$$\theta = -b_2\gamma_1 + \frac{2\gamma_1}{b_0 + b_1\lambda_1 + b_2\lambda_1^2}, \tag{7.20}$$

and where $\gamma_1, \gamma_2, b_0, b_1, b_2$ all retain their definitions given in Appendices A and B for the two mode coherent coupler system, and λ_1 remains the chosen root of (5.7). Let us then set the values of the conserved quantities in the degenerate FWM system (bar coordinates) in terms of parameters from the coherent coupler system (hat coordinates). To do this, we first define the constant d_5 in terms of λ_1 and the other d_j parameters from the coherent coupler system given in Appendix A such that:

$$d_5 = d_1 + 2d_2\lambda_1 + 3d_3\lambda_1^2 + 4d_4\lambda_1^3, \tag{7.21}$$

then we fix the following relationship betwen the intermodal power constants $\bar{\gamma}_j$ in the degenerate FWM system and the γ_k of the coherent coupler system such that:

$$\begin{aligned}\bar{u}_1\bar{v}_1 - \bar{u}_2\bar{v}_2 = \bar{\gamma}_1 - \bar{\gamma}_2 &= \frac{d_5\gamma_1}{4\left(\gamma_1^2 - \lambda_1^2\right)},\\ 2\bar{u}_1\bar{v}_1 - \bar{u}_3\bar{v}_3 = 2\bar{\gamma}_1 - \bar{\gamma}_3 &= \frac{d_5}{4\left(\gamma_1 - \lambda_1\right)},\\ 2\bar{u}_2\bar{v}_2 - \bar{u}_3\bar{v}_3 = 2\bar{\gamma}_2 - \bar{\gamma}_3 &= \frac{d_5}{4\left(\gamma_2 - \lambda_1\right)},\end{aligned} \tag{7.22}$$

where as always by definition:

$$\sum_{j=1}^{3}\bar{\gamma}_j = \sum_{j=1}^{2}\gamma_j = 0. \tag{7.23}$$

We then fix the free parameter χ in the degenerate FWM system in terms of parameters from the coherent coupler system such that:

$$\chi = \frac{8\left(b_0 + b_1\lambda_1 + b_2\lambda_1^2\right)}{d_5} \tag{7.24}$$

and the Hamiltonian in the degnerate FWM system takes the value:

$$\bar{H} = -\frac{b_2 d_5}{8} - \frac{2}{\chi} + \frac{64\left(d_5 + 16\lambda_1\right)\lambda_1}{\chi^3 d_5^2}. \tag{7.25}$$

With these paramater relationships the projection is fully defined and one can then verify that by differentiating (7.19) with respect to z, substituting the differential equations in (7.18) into the right hand side, then replacing the ratios of modes in bar coordinates for those in hat coordinates using (7.19), and simplifying using the Hamiltonian in (7.17), that one recovers the two mode coherent coupler system in (7.8). Thus it is established that there exists a projection from degenerate FWM to the coherent coupler.

The solutions to (7.18) are for $j = 1, 2, 3$:

$$\begin{aligned}\bar{u}_j(z) &= \sqrt{\frac{s\left(j\right)}{2}}\frac{\bar{\alpha}_j\sigma(z - 2z_0 + \mu_j)e^{z\left(\frac{\chi\bar{\gamma}_j}{s(j)} - \zeta(\mu_j - z_0) + \frac{1}{\chi}\right)}}{\sigma(\mu_j - z_0)\sigma(z - z_0)},\\ \bar{v}_j(z) &= \sqrt{\frac{s\left(j\right)}{2}}\frac{\sigma(z - \mu_j)e^{-z\left(\frac{\chi\bar{\gamma}_j}{s(j)} - \zeta(\mu_j - z_0) + \frac{1}{\chi}\right)}}{\bar{\alpha}_j\sigma(\mu_j - z_0)\sigma(z - z_0)},\\ \bar{u}_j(z)\bar{v}_j(z) &= \frac{s(j)}{2}\left(\wp(\mu_j - z_0) - \wp(z - z_0)\right)\end{aligned} \tag{7.26}$$

where $\bar{\alpha}_j$ is an integration constant that captures initial relative phase offsets between modes and can be used to equate initial conditions between the degenerate FWM and coherent coupler systems, $s(j)$ is the degeneracy weight such that $s(1) = s(2) = 1$ and $s(3) = 2$, the new third μ_j constant is $\mu_3 = z_1 + z_0$, the elliptic functions use invariants g_2, g_3, and $\mu_1, \mu_2, z_0, z_1, g_2, g_3$ share the same definitions and values as those given in Section 5 for the coherent coupler, thereby connecting the two systems.

8 Numerical validation and implementation

In this section we validate the analytic solutions through numerical evaluation and demonstrate their practical implementation using open-source software. This serves two purposes, first, to verify that the solutions correctly satisfy the differential equations, and second, to show that the Weierstrass elliptic functions can be readily evaluated using standard numerical libraries.

We evaluate the analytic solutions using the *pyweierstrass* Python package [7], which provides Weierstrass elliptic functions as a wrapper around Jacobi elliptic functions from the *mpmath* package [8]. For comparison, we solve the differential equations numerically using the DOP853 Runge-Kutta algorithm from *SciPy* [9].

Figures 1 and 2 show the real and imaginary parts of the modal power $u_j v_j$ using the analytic solution (5.15) compared against numerical integration of the abstract system (3.1). The analytic solutions (lines) are in excellent agreement with the numerical solutions (symbols). The intermodal power conservation laws in (4.7) are evident from the coordinated evolution of all four modes.

Figures 3 and 4 show the intensity and phase evolution of the physical field amplitudes A_j in a representative four-wave mixing scenario. The values were obtained by using (6.9) to calculate u_j, v_j, then converting to A_j, A_j^* via (**??**). Again, the analytic solutions (lines) match the numerical integration (symbols) throughout the propagation distance. The periodic energy exchange between modes visible in Figure 3 is characteristic of phase-matched four-wave mixing, whilst Figure 4 shows the phase evolution dominated by linear propagation with nonlinear contributions visible as regions of high curvature.

These comparisons confirm both the correctness of the analytic solutions and the feasibility of their numerical evaluation using readily available software tools.

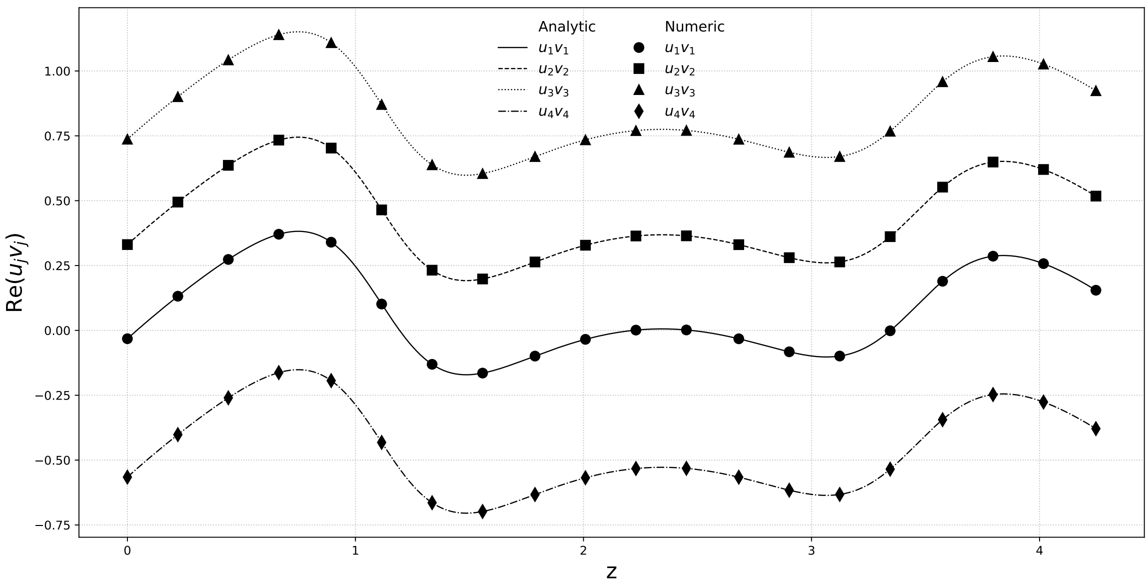

Figure 1: Real part of abstract modal power $u_j v_j$: analytic solution (5.15) (dashed lines) compared with numerical integration (symbols).

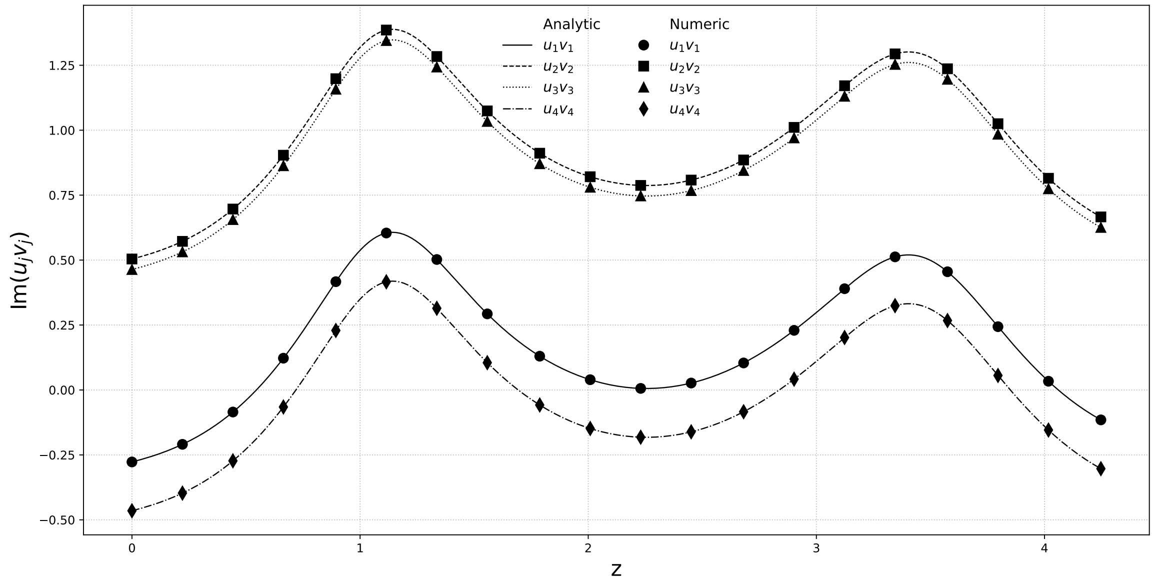

Figure 2: Imaginary part of abstract modal power u_jv_j: analytic solution (5.15) (dashed lines) compared with numerical integration (symbols).

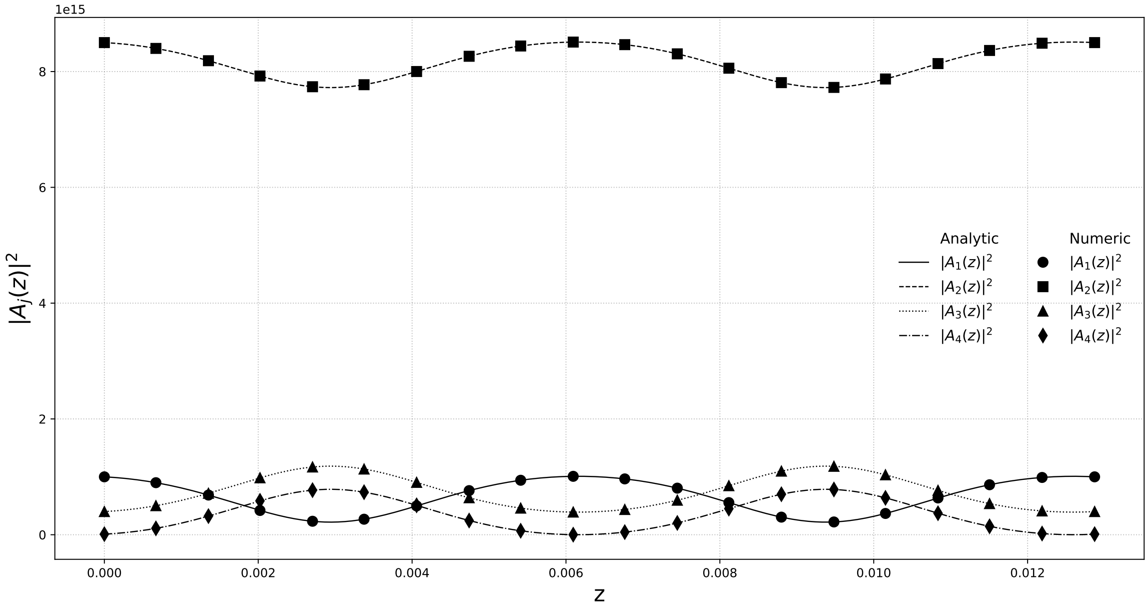

Figure 3: Intensity $|A_j|^2$ of physical field amplitudes: analytic solution (6.9) converted via (**??**) (dashed lines) compared with numerical integration (symbols).

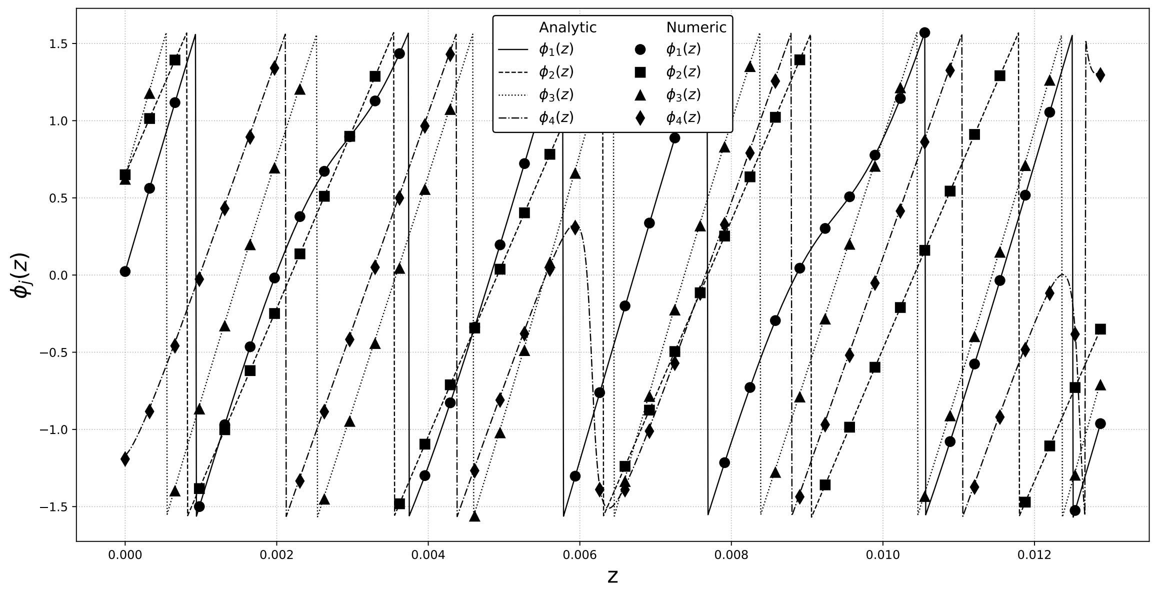

Figure 4: Phase ϕ_j of physical field amplitudes: analytic solution (6.9) converted via (**??**) (dashed lines) compared with numerical integration (symbols).

9 Conclusion

We have presented complete analytic solutions for quasi-continuous-wave four-wave mixing in nonlinear optical fibres without any of the standard simplifying assumptions. The solutions, expressed in terms of Weierstrass elliptic functions, provide the full complex field envelopes for all four waves under arbitrary initial conditions, including pump depletion, cross-phase modulation, and phase mismatch effects. These solutions complement previous work in the literature by providing a unified treatment of the complex envelopes themselves rather than treating amplitudes and phases separately.

Guided by the structure of these solutions, we developed a sequence of coordinate transformations that reveal a canonical form of the four-wave mixing equations. In these canonical coordinates, the system becomes parameter-free with a universal structure applying to all four-wave mixing processes regardless of specific fibre properties or operating conditions. The solutions in canonical coordinates are single-valued meromorphic Kronecker theta functions, connecting four-wave mixing to a broader class of integrable nonlinear optical systems. Remarkably, the transformations preserve the essential structure of the differential equations despite depending explicitly on the propagation variable, demonstrating a profound invariance property of four-wave mixing dynamics.

The connection between the Hamiltonian formulation and the Frobenius-Stickelberger determinant, established in Appendix C, provides deeper insight into why these conservation laws hold and links four-wave mixing to classical elliptic function theory. This mathematical framework may prove valuable for analysing related parametric processes and could extend to higher-order mixing or multimode interactions.

The methods developed here are broadly applicable to related nonlinear systems. In particular, the approach may be transferable to the analysis of mode coupling in multimode nonlinear fibre optics (see, e.g., Chapter 4 in [10]), where similar coupled-mode structures arise.

The practical utility of these solutions has been demonstrated through numerical evaluation using open-source Python libraries, confirming both their validity and computational accessibility. These analytic solutions may provide new insights for the design and optimisation of wavelength converters, parametric amplifiers, and quantum light sources by enabling exact analysis where approximations previously proved inadequate.

Acknowledgements

The author thanks Maria Dimou for a careful read of the manuscript.

Code Availability

The code used in this paper is available at [11] and includes Python Jupyter notebooks with symbolic derivations using *SymPy* [12] as well as the code for numeric validation used in Section 8.

A Parameter definitions in the original coordinates

This appendix provides explicit formulae for the parameters b_l, c_l, and d_l that appear in the quartic polynomial (5.5) and related expressions throughout Section 5. These parameters are expressed in terms of the system parameters a_j, $a_{j,k}$ and the conserved constants γ_j.

$$\begin{aligned}
N &= 2, && \text{(A.1)}\\
b_0 &= a_0 + \sum_{j=1}^{N} a_j\gamma_j + \frac{1}{2}\sum_{j,k=1}^{N} a_{j,k}\gamma_j\gamma_k,\\
b_1 &= -\sum_{j=1}^{N} a_j - \frac{1}{2}\sum_{j,k=1}^{N} (\gamma_j + \gamma_k)\, a_{j,k},\\
b_2 &= \frac{1}{2}\sum_{j,k=1}^{4} a_{j,k},\\
c_0 &= \prod_{j=1}^{N}\gamma_j,\\
c_1 &= 0,\\
c_2 &= 1,\\
d_0 &= b_0^2 - 4c_0,\\
d_1 &= 2b_0b_1,\\
d_2 &= 2b_0b_2 + b_1^2 - 4,\\
d_3 &= 2b_1b_2,\\
d_4 &= b_2^2 && \text{(A.2)}
\end{aligned}$$

B Initial condition relations

This appendix gives the relationships between initial conditions for the physical system and the constants γ_j and $\rho(0)$ used throughout the analysis. These relations enable determination of the constants from specified initial values of $u_j(0)$ and $v_j(0)$.

$$\begin{aligned}
N &= 2,\\
\rho(0) &= -\frac{1}{N}\sum_{j=1}^{N} u_j(0)v_j(0),\\
\frac{d}{dz}\rho(z)\Big|_{z=0} &= \prod_{j=1}^{N} u_j(0) - \prod_{j=1}^{N} v_j(0),\\
\gamma_j &= u_j(0)v_j(0) + \rho(0) && \text{(B.1)}
\end{aligned}$$

C The Frobenius-Stickelberger determinant

The Frobenius-Stickelberger (FS) determinant formula is an elliptic function identity relating products of σ functions to determinants of derivatives of $\wp$ [6, 13]. In this appendix we demonstrate that the conservation of the Hamiltonian can be viewed as a manifestation of this classical identity, providing a deeper structural understanding of why the Hamiltonian is conserved in the coherent coupler.

Our starting point is equation (C.4), which relates the wave mixing product to the phase modulation terms via the Hamiltonian. We will show that this equation can be transformed into the FS determinant formula by treating the left-hand side (the product of modes) and right-hand side (the polynomial in ρ) separately, each ultimately expressed in terms of Weierstrass functions.

We are interested in the 4×4 case of FS in the form:

$$\frac{C\,\sigma(z+\nu_1)\sigma(z+\nu_2)\sigma(z+\nu_3)\sigma(z-\nu_1-\nu_2-\nu_3)}{\sigma^4(z)} = \det M \tag{C.1}$$

where:

$$C = \frac{12\sigma(\nu_1-\nu_2)\sigma(\nu_1-\nu_3)\sigma(\nu_2-\nu_3)}{\sigma(\nu_1)\sigma^4(\nu_2)\sigma^4(\nu_3)} \tag{C.2}$$

$$\begin{aligned}\det M =& \begin{vmatrix} 1 & \wp(z) & \frac{\partial\wp(z)}{\partial z} & \frac{\partial^2\wp(z)}{\partial z^2} \\ 1 & \wp(\nu_1) & -\frac{\partial\wp(\nu_1)}{\partial\nu_1} & \frac{\partial^2\wp(\nu_1)}{\partial\nu_1^2} \\ 1 & \wp(\nu_2) & -\frac{\partial\wp(\nu_2)}{\partial\nu_2} & \frac{\partial^2\wp(\nu_2)}{\partial\nu_2^2} \\ 1 & \wp(\nu_3) & -\frac{\partial\wp(\nu_3)}{\partial\nu_3} & \frac{\partial^2\wp(\nu_3)}{\partial\nu_3^2} \end{vmatrix} \\ =& -6\left(\wp(\nu_1)-\wp(z)\right)\left(\wp(\nu_2)-\wp(z)\right)\left(\wp(\nu_1)-\wp(\nu_2)\right)\wp'(\nu_3) \\ &+6\left(\wp(\nu_1)-\wp(z)\right)\left(\wp(\nu_3)-\wp(z)\right)\left(\wp(\nu_1)-\wp(\nu_3)\right)\wp'(\nu_2) \\ &-6\left(\wp(\nu_2)-\wp(z)\right)\left(\wp(\nu_3)-\wp(z)\right)\left(\wp(\nu_2)-\wp(\nu_3)\right)\wp'(\nu_1) \\ &-6\left(\wp(\nu_1)-\wp(\nu_2)\right)\left(\wp(\nu_1)-\wp(\nu_3)\right)\left(\wp(\nu_2)-\wp(\nu_3)\right)\wp'(z)\end{aligned} \tag{C.3}$$

The starting point for us is the following relation established via (4.1), (4.5), and (5.2), in which any j applies in the RHS modal power derivative:

$$\begin{aligned}2\prod_{k=1}^{2}u_k =& a_0+\sum_{k=1}^{2}a_k u_k v_k+\frac{1}{2}\sum_{k,l=1}^{2}a_{k,l}u_k v_k u_l v_l-\frac{\partial}{\partial z}u_j\, v_j, \\ =& \rho'+\sum_{l=0}^{2}b_l\,\rho\,(z)^l\,.\end{aligned} \tag{C.4}$$

We will show how (C.4) can be transformed into (C.1)

C.1 The left hand side

From (5.15), the right hand side of (C.4) is a doubly periodic elliptic function, so too then is the left hand side. Thus for integers m, n and half-periods ω_1, ω_3 we must have:

$$\prod_{j=1}^{2}\frac{u_j(z)}{u_j(2m\omega_3+2n\omega_1+z)}=1. \tag{C.5}$$

We substitute our solution (6.9) into (C.5), take advantage of (6.13) to simplify, and use the quasi-periodicity of σ:

$$\sigma(2m\omega_3+2n\omega_1+z)=(-1)^{mn+m+n}\,\sigma(z)e^{(2m\omega_3+2n\omega_1+2z)\zeta(m\omega_3+n\omega_1)} \tag{C.6}$$

to reduce the product in (C.5) to an exponential which, as it is equal to one, must have argument $2\pi i\mathrm{N}(n,m)$, with $\mathrm{N}(n,m)$ an integer that varies with n, m. This leads to:

$$2i\pi\mathrm{N}(n,m)=-2\left(2z_1+\sum_{j=1}^{2}\nu_j\right)\zeta(m\omega_3+n\omega_1,g_2,g_3)-(2m\omega_3+2n\omega_1)\sum_{j=1}^{2}r_{0,j}, \tag{C.7}$$

$$2z_1+\sum_{j=1}^{2}\nu_j=-2\mathrm{N}(0,1)\omega_1+2\mathrm{N}(1,0)\omega_3=0 \quad (\text{mod lattice}), \tag{C.8}$$

$$\sum_{j=1}^{2}r_{0,j}=-2\zeta(m\omega_3+n\omega_1,g_2,g_3) \tag{C.9}$$

where $\nu_j=\mu_j-z_0$, and in moving from (C.7) to (C.8) and (C.9) we made use of the identity $\zeta(\omega_3)\omega_1-\zeta(\omega_1)\omega_3=i\pi/2$. Ultimately, equations (C.6), (C.8) and (C.9) allow us to make the following substitution (or the equivalent at $z\to z-z_0$) in the left hand side of (C.4):

$$\frac{\sigma(z+z_1)}{\sigma(z_1)}\exp\left(z\sum_{j=1}^{2}r_{0,j}\right)=-\frac{\sigma(z-z_1-\nu_1-\nu_2)}{\sigma(z_1+\nu_1+\nu_2)}. \tag{C.10}$$

C.2 The right hand side

The following polynomial interpolation formula holds for any values of the variables and functions:

$$\begin{aligned}\sum_{l=0}^{2} b_l\,\rho\,(z)^l = \frac{\gamma_1-\rho(z)}{\gamma_1-\gamma_2}\sum_{l=0}^{2} b_l\,\gamma_1^l + \\ \frac{\gamma_2-\rho(z)}{\gamma_2-\gamma_1}\sum_{l=0}^{2} b_l\,\gamma_2^l + \\ \left(\gamma_1-\rho(z)\right)\left(\gamma_2-\rho(z)\right) b_2\end{aligned} \tag{C.11}$$

where in our case $\gamma_j = \rho(\mu_j)$ and the sum over $b_l\gamma_j$ can be written using (5.4) as:

$$\sum_{l=0}^{2} b_l\,\gamma_j^l = \rho'\left(\mu_j\right). \tag{C.12}$$

We can then use (5.15) to express ρ in terms of $\wp$ so that we can substitute all variables in (C.11) for values in terms of $\wp, \wp'$. We then divide both sides of (C.4) by:

$$\frac{\wp'(z_1)}{b_2\left(\wp(z_1)-\wp(z-z_0)\right)^2} \tag{C.13}$$

and observe that it takes the equivalent form to (C.1) up to constant factors that may look different but can be shown to be identical by evaluating at $z \to z + z_0$, multiplying by $\sigma(z)^4$, and taking $z \to 0$.

This completes the demonstration that the Hamiltonian conservation law (C.4) is equivalent to the Frobenius-Stickelberger determinant formula. This connection reveals that the conserved Hamiltonian of the coherent coupler is not merely a convenient construct but rather a manifestation of deep structural properties of elliptic functions. The FS determinant framework may provide tools for generalizing these results to higher-order wave mixing processes or systems with additional degrees of freedom.

D Parameter definitions in the bar coordinates

This appendix provides explicit formulae for the parameters that appear in the bar coordinate system introduced in Section **??**, including the key transformation parameters d_5, ς, χ, and the rescaled constants $\bar{\gamma}_j$, θ_j, and $\bar{\kappa}_j$.

$$\begin{aligned}
d_5 &= d_1 + 2d_2\lambda_1 + 3d_3\lambda_1^2 + 4d_4\lambda_1^3, \\
\varsigma &= \frac{1}{2}\left(b_0 + b_1\lambda_1 + b_2\lambda_1^2\right)\left(\sqrt{\prod_{j=1}^{4}\gamma_j - \lambda_1}\right)^{-1} = \pm 1, && \text{(D.1)} \\
\chi &= \frac{8\left(b_0 + b_1\lambda_1 + b_2\lambda_1^2\right)}{d_5} = \frac{16\,\varsigma}{d_5}\sqrt{\prod_{j=1}^{4}\gamma_j - \lambda_1}, \\
\bar{\gamma}_j &= \frac{d_5}{4\left(\gamma_j - \lambda_1\right)} + \frac{b_1 + 2b_2\lambda_1}{\chi} - \frac{4}{\chi^2}, \\
\theta_j &= \frac{b_0 + b_1\lambda_1 + b_2\lambda_1^2}{2\left(\gamma_j - \lambda_1\right)} + \frac{b_1}{4} + \frac{b_2\gamma_j}{2} + \frac{b_2\lambda_1}{2} - \frac{1}{\chi}, \quad \sum_{j=1}^{4}\theta_j = 0, \\
\bar{\kappa}_j &= 2\theta_j + \frac{1}{\chi} - b_2\gamma_j - \zeta(\mu_j - z_0, g_2, g_3). && \text{(D.2)}
\end{aligned}$$

E Scaled parameters in the tilde coordinates

When the system in subsection **??** (tilde notation) is considered a transform of that in subsection **??** (bar notation), then parameters are related via the following scaling laws:

$$\begin{aligned}\tilde{H} &= \chi^3 \bar{H}, & \text{(E.1)}\\ \tilde{\epsilon}_j &= \epsilon_j \sqrt[4]{\varsigma},\\ \tilde{\gamma}_j &= \chi^2 \bar{\gamma}_j,\\ \tilde{g}_2 &= \chi^4 g_2,\\ \tilde{g}_3 &= \chi^6 g_3,\\ \xi_0 &= \frac{z_0}{\chi},\\ \tilde{\mu}_j &= \frac{\mu_j}{\chi}. & \text{(E.2)}\end{aligned}$$

If the system in subsection **??** is considered as a standalone system irrespective of any transformation, then the paramters can be given exclusively in the coordinate system denoted by tilde notation as follows. Firstly, $\tilde{H}$ can be obtained from substituting $\tilde{u}_j(0), \tilde{v}_j(0)$ into (**??**) and $\tilde{\epsilon}_j$ can be considered an integration constant fixed by initial conditions to capture any phase offset between $\tilde{u}_j, \tilde{v}_j$. We may further use relations:

$$\begin{aligned}\tilde{\gamma}_j =& \tilde{u}_j(0)\tilde{v}_j(0) - \frac{1}{4}\sum_{j=1}^{4} \tilde{u}_j(0)\tilde{v}_j(0),\\ \tilde{g}_2 =& \frac{\tilde{H}^2}{12} + \tilde{H}\left(\frac{\tilde{e}_2}{12} + \frac{16}{3}\right) + \frac{\tilde{e}_2^2}{48} + \frac{2\tilde{e}_2}{3} - \frac{\tilde{e}_3}{4} + \frac{64}{3},\\ \tilde{g}_3 =& \frac{\tilde{H}^3}{216} + \tilde{H}^2\left(\frac{\tilde{e}_2}{144} - \frac{5}{9}\right) + \tilde{H}\left(\frac{\tilde{e}_2^2}{288} - \frac{2\tilde{e}_2}{9} - \frac{\tilde{e}_3}{48} - \frac{64}{9}\right)\\ &+ \frac{\tilde{e}_2^3}{1728} - \frac{5\tilde{e}_2^2}{144} - \frac{\tilde{e}_2\tilde{e}_3}{96} - \frac{8\tilde{e}_2}{9} + \frac{\tilde{e}_3}{3} + \frac{\tilde{e}_4}{4} - \frac{512}{27} \qquad \text{(E.3)}\end{aligned}$$

where $\tilde{e}_j = \tilde{e}_j\left(\tilde{\gamma}_1, \tilde{\gamma}_2, \tilde{\gamma}_3, \tilde{\gamma}_4\right)$ is the j^{th} order *elementary symmetric polynomial* formed with the four $\tilde{\gamma}_j$. The points $\xi_0, \tilde{\mu}_j$ are found by inverting:

$$\begin{aligned}\wp(\xi_0, \tilde{g}_2, \tilde{g}_3) &= -\frac{\tilde{H}}{12} - \frac{\tilde{e}_2}{24} - \frac{\sum_{j=1}^{4} \tilde{u}(0,\mu_j)\tilde{v}(0,\mu_j)}{4} + \frac{4}{3},\\ \wp'(\xi_0, \tilde{g}_2, \tilde{g}_3) &= \frac{1}{4}\left(\prod_{j=1}^{4} \tilde{u}_j(0) - \prod_{j=1}^{4} \tilde{v}_j(0)\right),\\ \wp(\tilde{\mu}_j - \xi_0, \tilde{g}_2, \tilde{g}_3) &= -\frac{\tilde{H}}{12} - \frac{\tilde{e}_2}{24} + \tilde{\gamma}_j + \frac{4}{3},\\ \wp'(\tilde{\mu}_j - \xi_0, \tilde{g}_2, \tilde{g}_3) &= \tilde{H} + \frac{\tilde{e}_2}{4} - \frac{\tilde{\gamma}_j^2}{2} - 4\tilde{\gamma}_j. & \text{(E.4)}\end{aligned}$$

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